

Module 9 Cisco IT Essentials Study Guide

This study guide focuses on key concepts and terms related to virtualization, cloud computing, and hypervisors, helping students prepare for the quiz.

Key Concepts

Virtualization

- Virtualization is the process of creating a virtual version of something, such as hardware platforms, storage devices, or network resources. It is a foundational concept in cloud computing.
- Virtual machines (VMs) run their own operating systems and can coexist on a single physical machine, sharing its resources.

Cloud Computing

- Cloud computing provides access to shared computing resources (e.g., servers, storage, applications) over the Internet. It allows businesses to use services on-demand without owning physical infrastructure.
- Cloud computing separates applications from hardware, enabling scalability and flexibility.
- Examples of cloud-based services include Gmail, Google Drive, OneDrive, Outlook Online, and Dropbox.

Hypervisors

- Hypervisors are software or hardware solutions that enable virtualization by allowing multiple virtual machines to run on a single physical host.
 - **Type 1 Hypervisor (Bare Metal):** Installed directly on hardware for maximum efficiency and direct access to hardware resources. Examples include VMware vSphere and Windows 10 Hyper-V.
 - **Type 2 Hypervisor (Hosted):** Runs on top of an operating system. Examples include Microsoft Virtual PC and Oracle VM VirtualBox.

Benefits of Virtualization

- Increased efficiency: Virtualized servers can optimize resource usage.

- Direct access to hardware resources: Bare metal hypervisors provide better performance compared to hosted hypervisors.

Cloud Service Models

1. **Software as a Service (SaaS):** Provides software applications over the Internet (e.g., payroll systems). Users access applications without managing infrastructure or platforms.
2. **Infrastructure as a Service (IaaS):** Offers virtualized computing resources like network hardware (routers, switches) over the Internet.
3. **IT as a Service (ITaaS):** Outsourced IT services including training, software licensing, and provisioning.
4. **Platform as a Service (PaaS):** Provides a platform for developers to build applications without worrying about underlying infrastructure.

System Requirements for Virtualization

- To run Windows 10 or 11 Hyper-V virtualization platform:
 - Minimum RAM requirement: 4 GB
 - Recommended RAM for better performance when running multiple VMs: More than 4 GB

Vocabulary

1. **Virtualization:** The creation of virtual versions of physical components like servers or networks.
2. **Cloud Computing:** A model that delivers computing resources and services over the Internet on-demand.
3. **Hypervisor:** Software or hardware that allows multiple virtual machines to run simultaneously on a single host machine.
 - **Type 1 Hypervisor:** Bare metal hypervisor installed directly on hardware.
 - **Type 2 Hypervisor:** Hosted hypervisor running on top of an operating system.
4. **VM (Virtual Machine):** A software-based emulation of a physical computer that runs its own operating system.
5. **SaaS (Software as a Service):** A cloud service model providing software applications over the Internet without requiring local installation.

6. **IaaS (Infrastructure as a Service):** A cloud service model offering virtualized computing resources like servers and network hardware.
7. **ITaaS (IT as a Service):** Outsourced IT solutions including user training, provisioning, and licensing.
8. **RAM (Random Access Memory):** A type of computer memory required for running operating systems and applications efficiently in virtualized environments.
9. **OneDrive:** A cloud storage service provided by Microsoft for storing files online.
10. **Google Drive:** A cloud storage service offered by Google for file storage and collaboration.
11. **Outlook Online:** A web-based email service provided by Microsoft as part of its SaaS offerings.